

REthinking the Welfare **STATE** under Political Constraints in a Changing World

Sustainable Social Security

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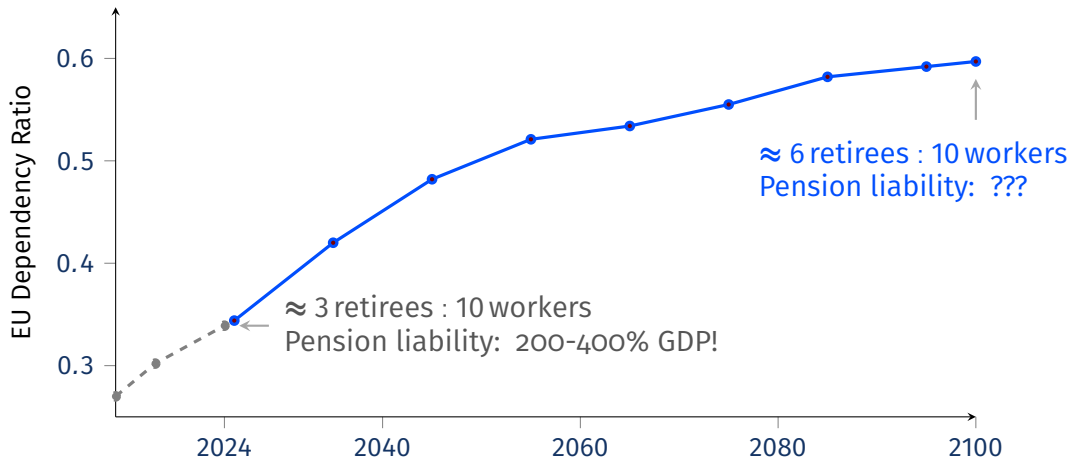
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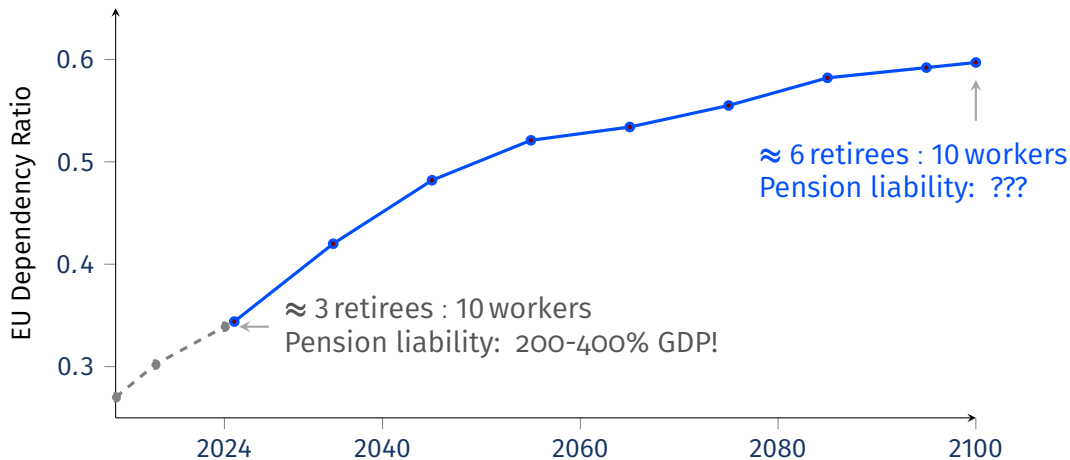
RESTATE

A Urgent Problem



Demographic Changes ⇒ Fiscal Sustainability

A Urgent Problem



Demographic Changes ⇒ Fiscal Sustainability vs. Political Implementability

A Big Beautiful Question

How should the welfare state be designed to balance
fiscal sustainability and **political implementability**
in a changing world?

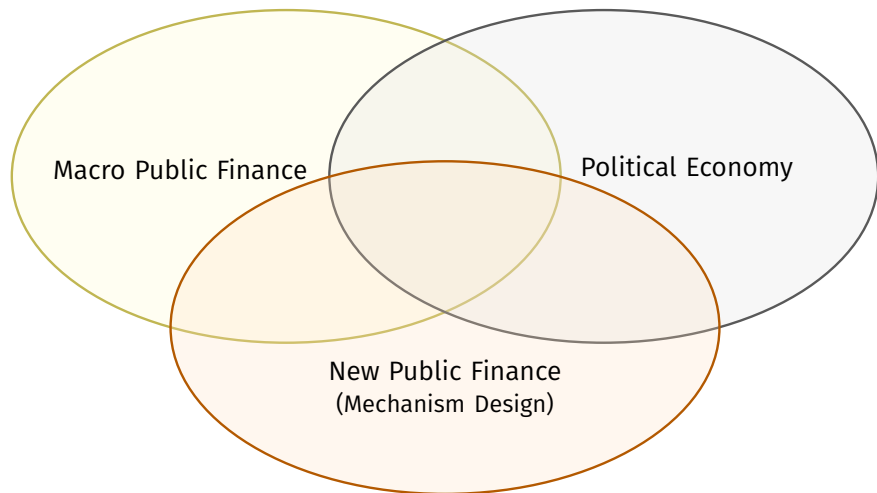
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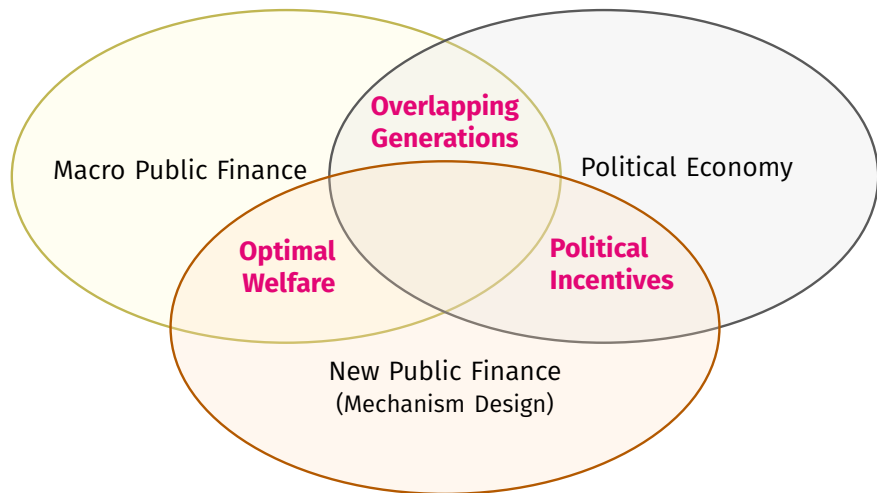
⇒ **First-best policies** are **unattainable**

⇒ **Intergenerational redistribution** is **key**

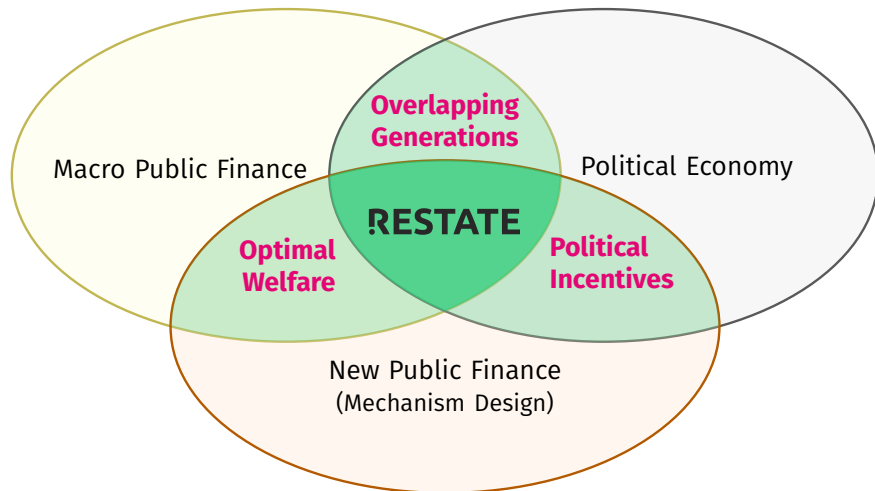
Need for a New Framework



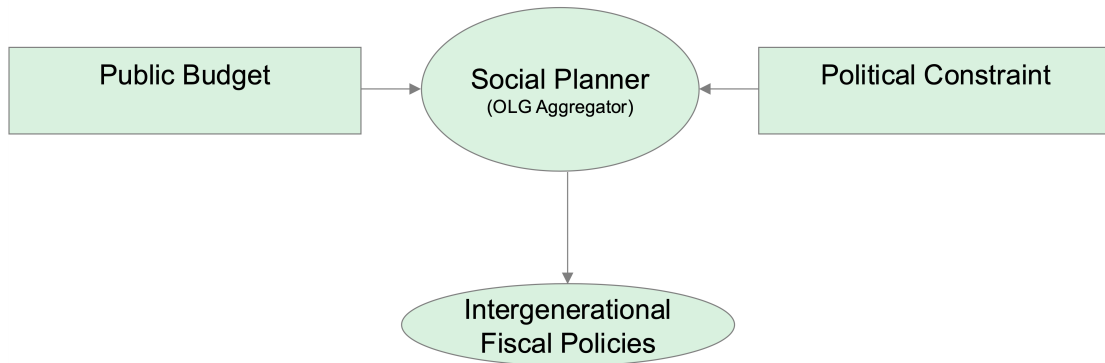
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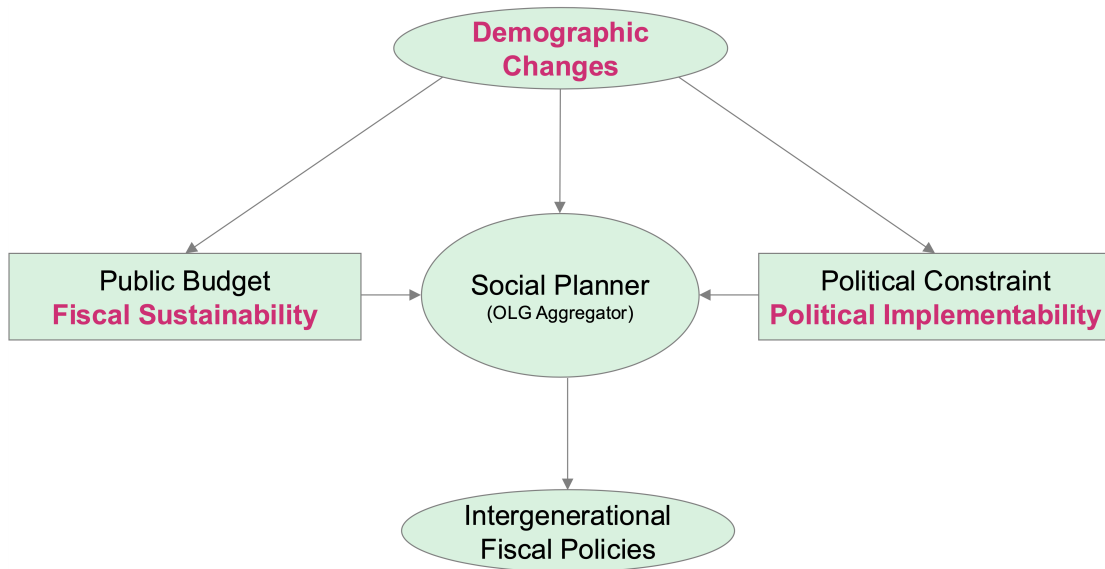
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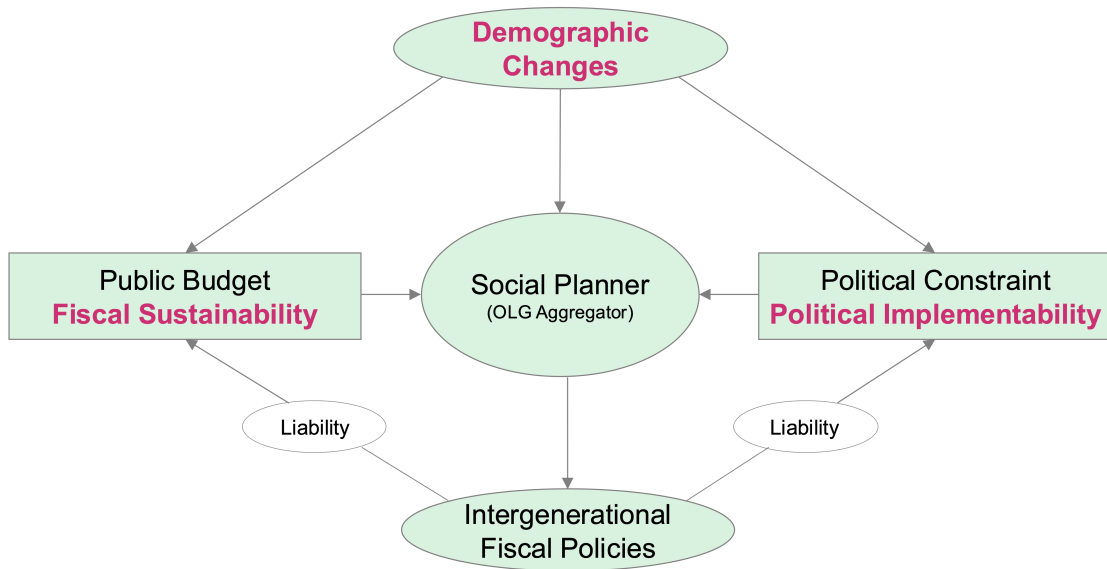
Main Building Blocks



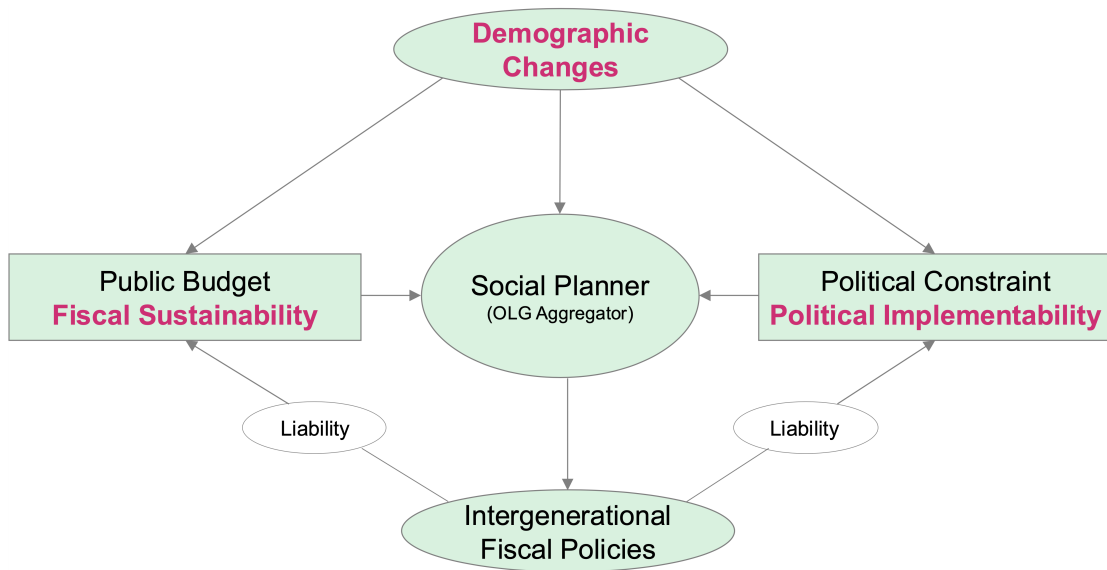
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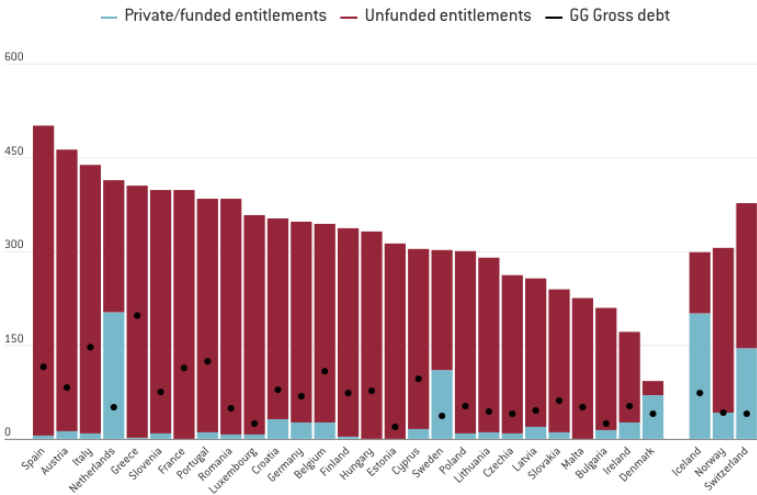


Fiscal Inertia — **Nonlinear Policies** — **Inequality**

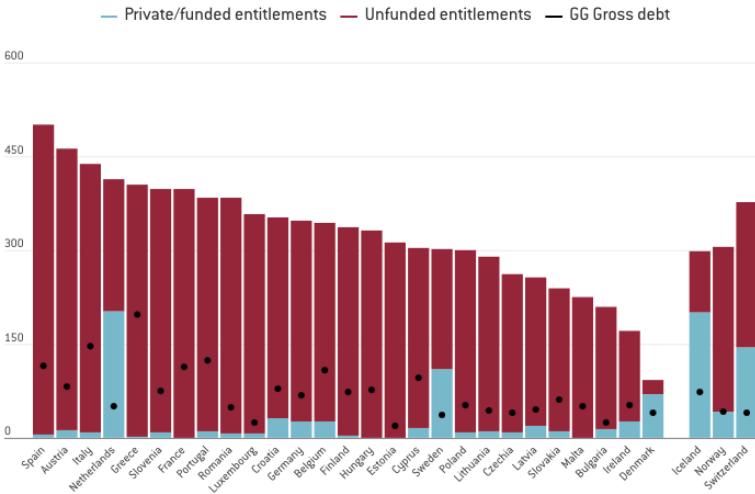
Sustainable Social Security

with S. Cappellini (Navarra), F. Lancia (Ca' Foscari Venice), T. Worrall (Edinburgh)

Accrued Social Security Promises



Accrued Social Security Promises



Sustainable Social Security

is challenged by rising

Dependency Ratio

and

Demographic Uncertainty

Political Constraints in Reality



Sustainable Social Security is challenged by **Limited Enforcement**:

Old demand maintenance of pension promises vs

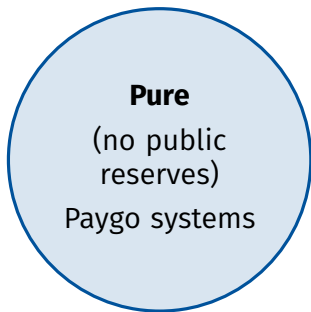
Young oppose a higher fiscal burden.

How should a **Sustainable Social Security** be structured if
Limited Enforcement and **Demographic Uncertainty**?

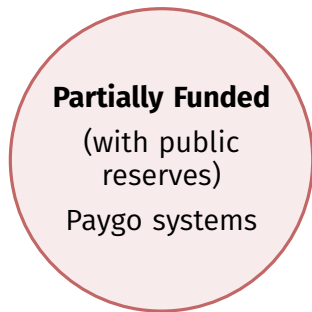
What **rule** should **link pension benefits** and **contributions**?

How should a **Sustainable Social Security** be structured if
Limited Enforcement and **Demographic Uncertainty**?

What **rule** should **link pension benefits** and **contributions**?



vs.



Existing Approaches

Social security design

under full mandate and info frictions

Demange & Laroque (1999); Conesa & Garriga (2008);
Hosseini & Shourideh (2019)

Social security as a voting outcome of a dynamic game

Cooley & Soares (1999); Azariadis & Galasso (2002);
Bassetto (2008); Gonzalez-Eiras & Niepelt (2008)

Optimal insurance **under limited enforcement**

Thomas & Worrall (1988); Kocherlakota (1996); Abraham
& Laczó (2018); Lancia, Russo & Worrall (2024)

This Approach

→ Considers **limited enforcement**
as a form of political frictions

→ Characterizes **Pareto optimal**
rule, implementable as a voting
outcome

→ OLG setting with **aggregate**
demographic risk and **public**
savings

A Simple Framework

Demographics

- Two-period OLG: **Young** (N_t^y), **Old** (N_t^o)
- **Stochastic survival**: $\phi_i \in \{\phi_L, \phi_H\}$, $\phi_L < \phi_H$, π_{ij} transition prob.
- **Dependency ratio**: $\frac{N_t^o}{N_t^y} = \phi_t$

Economy

- Young endowed with w and Old with α
- Discount factor $\beta \in (0, 1]$, time-separable intertemporal utility
- In autarky, **dynamic inefficiency**
- Young pay **contributions** τ_t to finance **pensions** p_t

Preferences

$$u(w(1 - \tau_t)) + \beta \mathbb{E}_t[\phi_{t+1} u(\alpha + p_{t+1})]$$

Public Budget

In each period, the public budget is:

$$\text{No Fund : } \underbrace{\tau_t \cdot w \cdot N_t^y}_{\text{contributions}} = \underbrace{p_t \cdot N_t^o}_{\text{pensions}} \quad \Rightarrow \quad p_t = \frac{1}{\phi_t} \cdot w \tau_t$$

Pensions are tied one-for-one to current contributions and demographics.

$$\begin{aligned} \text{With Fund : } \underbrace{\tau_t w - \phi_t p_t}_{\text{surplus}} &= \underbrace{q F_{t+1} - F_t}_{\text{net asset position}} \quad \text{with } F_{t+1} \geq 0 \\ \Rightarrow \underbrace{\tau_t w - q F_{t+1}}_{\text{net revenue}} &= \underbrace{\phi_t p_t - F_t}_{\text{net liability}} \end{aligned}$$

*The fund decouples pensions from current demographics:
surpluses build assets, deficits draw them down.*

Social Planner Problem

A **benevolent planner** maximizes the sum of expected discounted utility of **all** generations, weighting future generations by $\delta \in (0, 1)$:

Intergenerational Pareto Frontier For $i \in \{L, H\}$:

$$V_i(p, F) = \max_{p^\dagger, \tau, (p'_j)_j, F'} \left\{ \underbrace{\frac{\beta}{\delta} \cdot \phi_i}_{\text{Old's weight}} \underbrace{u(\alpha + p^\dagger)}_{\text{Old's payoff}} + \underbrace{u(w(1 - \tau))}_{\text{Young's payoff}} + \underbrace{\delta \sum_j \pi_{ij} V_j(p'_j, F')}_{\text{Expected future value}} \right\}$$

$$\text{with } p^\dagger = \frac{1}{\phi}(\tau w + F - qF') \Rightarrow \text{Pension paid}$$

$p \Rightarrow$ Pension promised

Political Sustainability Constraints

Promises to Old must be honored $\Rightarrow$ **Promise-Keeping (PK) Constraint**

$$u(\alpha + p^+) \geq u(\alpha + p)$$

Multiplier: $\frac{\beta}{\delta} \cdot \phi_i \cdot \lambda$

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Young are willing to participate $\Rightarrow$ **Participation Constraint (PC)**

$$\underbrace{u(w(1-\tau)) - u(w)}_{\text{Young's sacrifice}} + \beta \sum_j \pi_{ij} \phi_j \underbrace{[u(\alpha + p_j) - u(\alpha)]}_{\text{Old's gain}} \geq 0$$

Multiplier: μ

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Multiplier: μ

Policy must balance **credibility (PK)** and **incentives (PC)**

Political Sustainability Constraints

A more general formulation of **Participation Constraint (PC)**:

$$\underbrace{\omega v^y(\tau, \{p'_j\})}_{\text{young}} + \underbrace{(1-\omega) v^o(p^\dagger)}_{\text{old}} \geq \underline{v}(p, F)$$

Solving the Social Planner's Problem

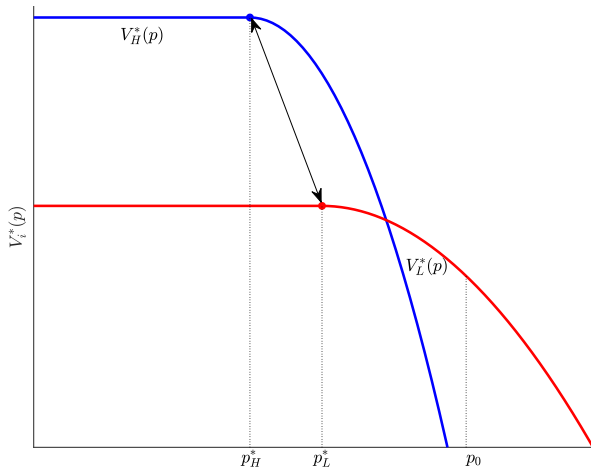
The goal is to achieve the **first best** for **optimal consumption redistribution**

$$\frac{u_c(w(1-\tau))}{u_c(\alpha+p)} = \frac{\beta}{\delta}$$

with $F' = 0$ if high q

⇒ **Defined Benefit System**

$p_L^* > p_H^*$ independent of contributions



Solving the Social Planner's Problem

Sometimes the first best is unattainable

⇒ High dependency ratio (high ϕ) or high accrued promises (high p)

⇒ **Redistribution wedges:**

Solving the Social Planner's Problem

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$$\frac{u_c(w(1-\tau))}{u_c(\alpha+p)} = \frac{\beta}{\delta} \cdot \frac{\overbrace{1+\lambda}^{\text{PK binds} \approx \text{Old weight}}}{\underbrace{1+\mu}_{\text{PC binds} \approx \text{Young weight}}}$$

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↑
**Endogenous
Generations Weight**

Solving the Social Planner's Problem

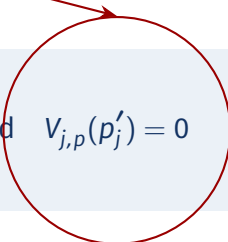
Low Dependency Ratio $\Rightarrow$ *PC Slack*: $\mu = 0$

$$\frac{u_c(w(1-\tau))}{u_c(\alpha+p)} = \frac{\beta}{\delta} \cdot (1 + \lambda) \quad \text{and} \quad v_{j,p}(p'_j) = 0$$

Solving the Social Planner's Problem

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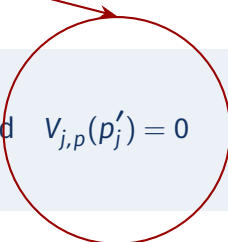
Future pension promises reset


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Promised pension set to undistorted level

taxes are collected to pay the current old

$\rightarrow$ **Defined Benefit Component**

Solving the Social Planner's Problem

High Dependency Ratio $\Rightarrow$ *PC Binds*: $\mu > 0$

$$\frac{u_c(w(1-\tau))}{u_c(\alpha+p)} = \frac{\beta}{\delta} \cdot \frac{(1+\lambda)}{(1+\mu)} \quad \text{and} \quad V_{j,p}(p'_j) \approx -\mu < 0$$

Solving the Social Planner's Problem

High Dependency Ratio $\Rightarrow$ PC Binds: $\mu > 0$

Future pension promises
rolled over

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Updating Rule : $\underbrace{\mu}_{\text{Young weight today}} = \underbrace{\lambda'}_{\text{Old weight tomorrow}} \Rightarrow \text{Fiscal Inertia}$

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Promised pension has to go up to keep the young happy

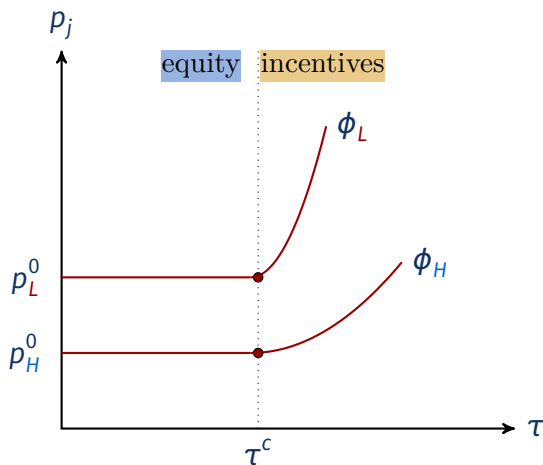
Benefits are dependent on contributions paid

$\rightarrow$ **Contribution-based Component**

Social Security Rule

Nonlinear optimal social-security rule linking pension benefits to contributions.

$$p'_j \approx \begin{cases} p_{j,\min} + \theta f_j(w\tau) & \text{if } \tau > \tau^c \\ p_{j,\min} & \text{if } \tau \leq \tau^c \end{cases}$$

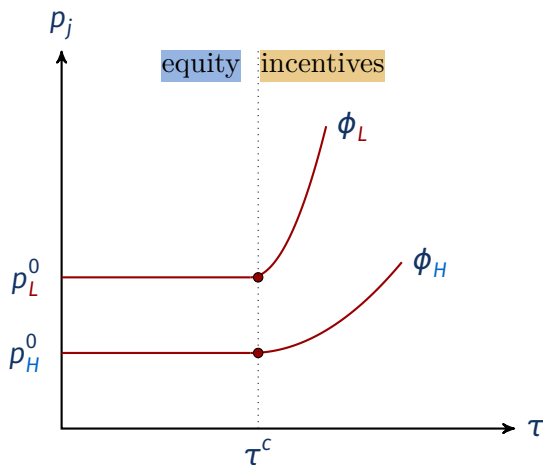


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Minimum
pension



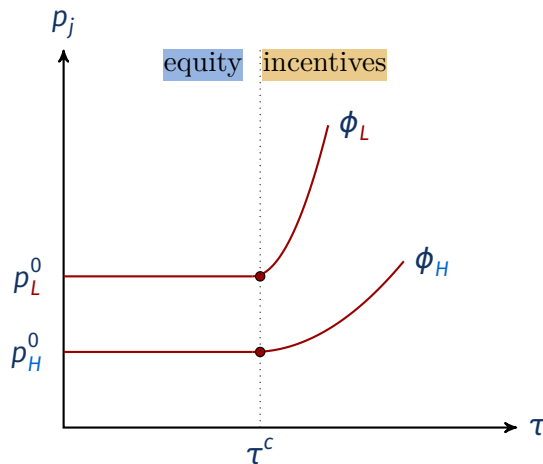
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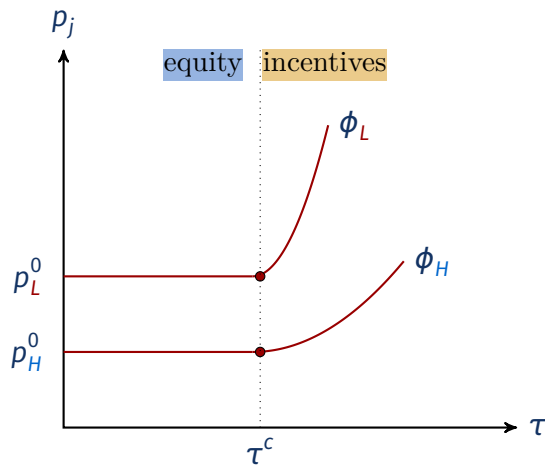
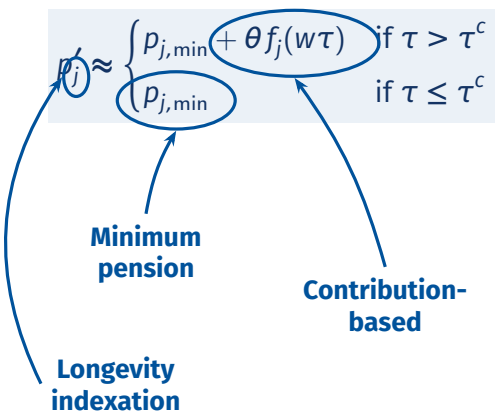
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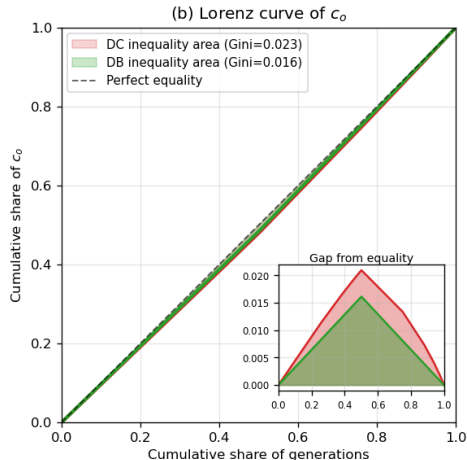
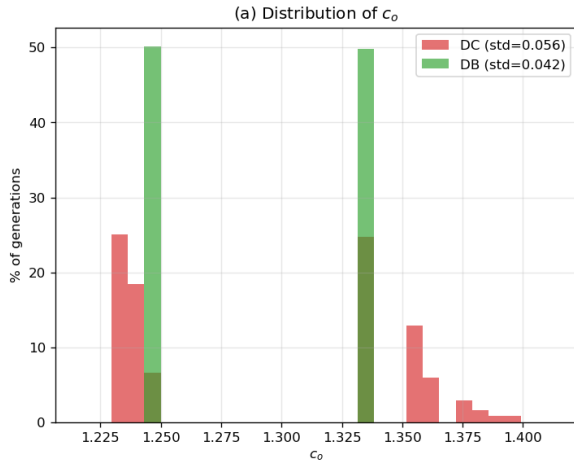


Social Security Rule

Nonlinear optimal social-security rule linking pension benefits to contributions.



Social Security Rule



Numerical Illustration: iid demographic shocks with $\pi_{ij} = 1/2$ and $i = H, L$

Intergenerational **inequality** from the equity–incentives tradeoff
in sustainable social security.

Social Security with Public Fund

Full Enforcement

Link via ***Euler condition***



- **Precautionary saving:** build up in ϕ_L , draw down in $\phi_H \Rightarrow$ consumption smoothing
- **No accumulation** ($F' = 0$) when q is large enough

Social Security with Public Fund

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Limited Enforcement

Link via **Euler + updating condition**



- **Fund overaccumulation**: assets relax future participation constraints $\Rightarrow$ new precautionary motive

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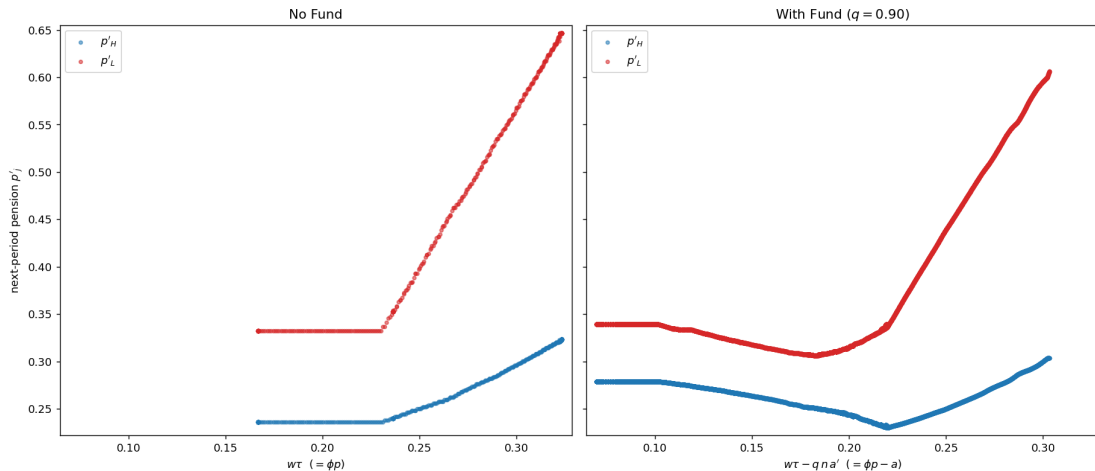
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Policy: a Defined Benefit scheme is more sustainable in the presence of a public fund.

Social Security Rule With Fund



Takeaway

Optimal policies under political constraints (enforcement limits)
are **endogenous functions of underlying states**:

fiscal inertia + non-linearity + inequality

Why Relevant

“Economics is one input to the design of [fiscal] rules, but as a practical matter, without fully accounting for political incentives, economically determined rules are pointless: they have little chance of being implemented; if adopted, they are unlikely to be obeyed.”

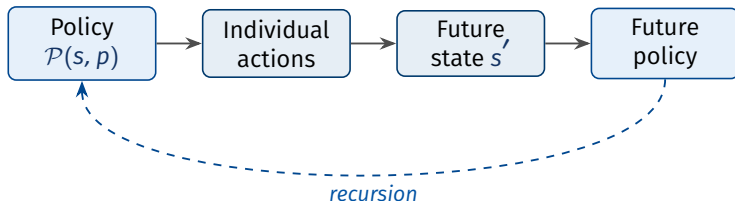
— Eric Leeper (2023)

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Theoretical and Quantitative Implications



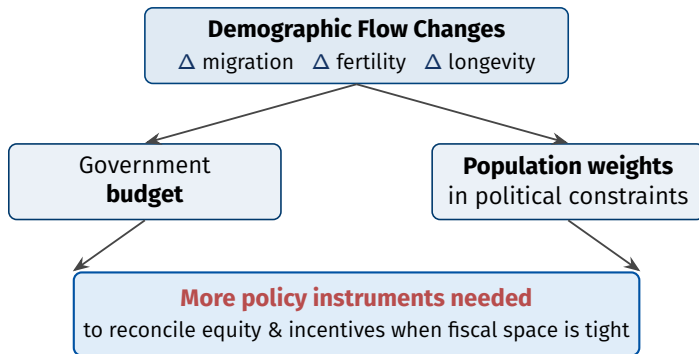
- Explains **persistence** of dynamically inefficient policies
- Rationalizes **asymmetric** entrenchment / retrenchment reforms

Takeaway

Optimal policies under political constraints (enforcement limits)
are **endogenous functions of underlying states**:

fiscal inertia + non-linearity + inequality

Theoretical and Quantitative Implications



Fiscal policy is not just accounting
It's political device for long-run sustainable Welfare State

<https://alessiarussoecon.weebly.com/erc—restate.html>

First Best

If q high that $a' = 0$, with CRRA utility and risk aversion coefficient γ :

$$p^*(\phi) = \frac{w - \alpha \left(\frac{\delta}{\beta}\right)^{\frac{1}{\gamma}}}{\phi + \left(\frac{\delta}{\beta}\right)^{\frac{1}{\gamma}}} \quad \text{and} \quad \tau^*(\phi) = \frac{\phi \left(w - \alpha \left(\frac{\delta}{\beta}\right)^{\frac{1}{\gamma}} \right)}{w \left(\phi + \left(\frac{\delta}{\beta}\right)^{\frac{1}{\gamma}} \right)}$$

$\Rightarrow$ An increase in ϕ reduces $p^*(\phi)$ and raises $\tau^*(\phi)$

$\Rightarrow$ Pension promises do not depend on the contributions paid

When $\beta = \delta$, $c^{o*}(\phi) = c^{y*}(\phi) = \frac{w + \alpha\phi}{1 + \phi}$ for any ϕ

When $\phi_i < \phi_j$, $c_i^{y*} > c_j^{y*}$ and $c_i^{o*} > c_j^{o*}$

With Asset Accumulation

If q is not too high, agents have incentives to accumulate assets:

$$u'(\alpha + p) = \frac{\delta}{q} \sum_j \pi_{ij} u'(\alpha + p'_j)$$

- Non-state-contingent asset $\Rightarrow$ **incomplete insurance** across states j
- **Precautionary saving** from two channels:
 - *Prudence* ($u''' > 0$): convexity of u' raises expected marginal utility
 - *Borrowing constraints*: bind in high- ϕ states, amplifying saving motive

Benchmark: No PC

PK does *not* bind

Constant ratio of marginal utilities:

$$\frac{u'(w(1-\tau))}{u'(\alpha+p)} = \frac{\beta}{\delta}$$

First-best policy:

- $p_H^* < p_L^*$ and $\tau_H^* > \tau_L^*$
- Perfect, history-independent risk sharing

Benchmark: No PC

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PK binds

Wedge distorts the ratio above β/δ :

$$\frac{u'(w(1-\tau))}{u'(\alpha+p)} = \frac{\beta}{\delta} (1 + \lambda(p, \phi_i))$$

Distortion is **temporary** (at most one period):

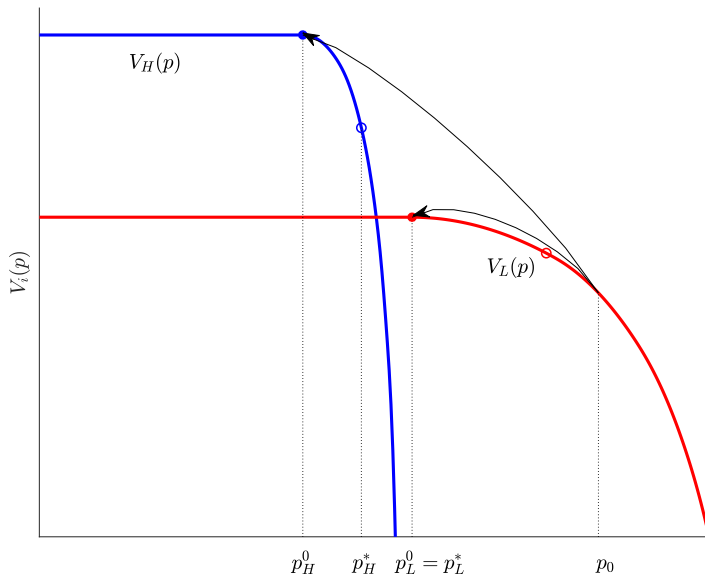
$$V'_j(p'_j) = 0 \Rightarrow p_j^* = \sup \{p \mid V'_j(p) = 0\}$$

$$V'_i(p) = -\frac{\beta}{\delta} \cdot \phi_i \cdot u'(\alpha+p) \cdot \lambda(p, \phi_i)$$

$$\Rightarrow \lambda(p'_j, \phi_j) = 0 \quad \forall j$$

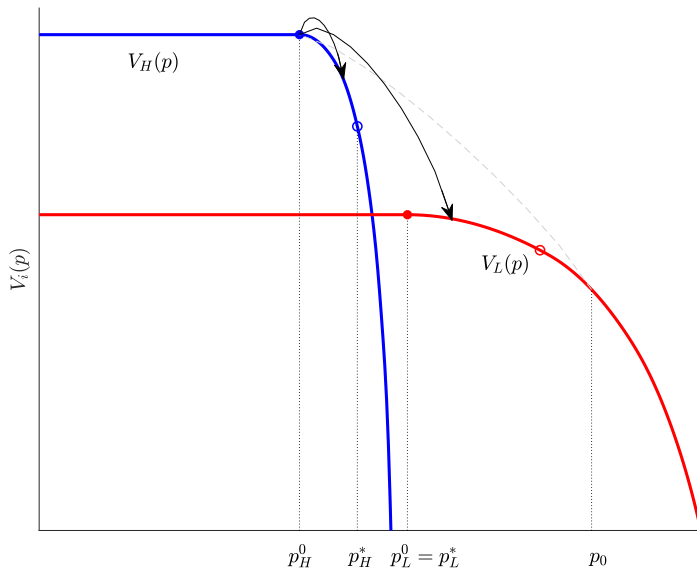
Low Dependency Ratio $\Rightarrow$ PC Slack

Start from $\phi_0 = \phi_L$ and $p_0 > p_L^0$



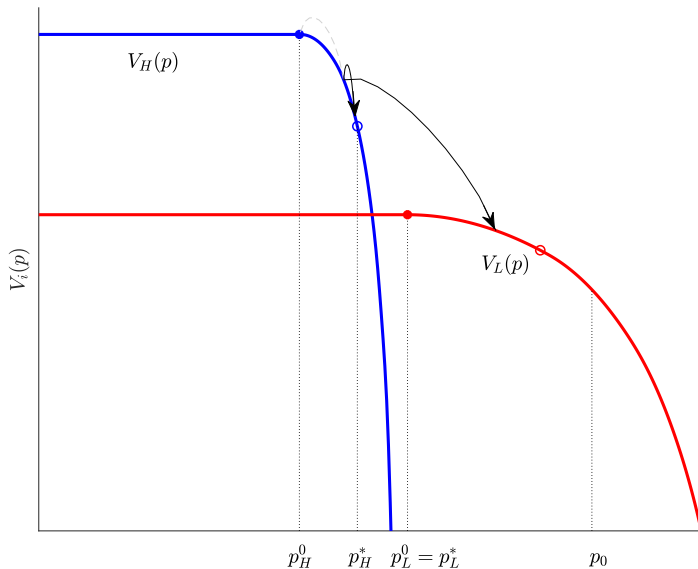
High Dependency Ratio $\Rightarrow$ PC Binds

If $\phi_1 = \phi_H$, promised pensions must be larger than p_j^0 to relax participation



High Dependency Ratio $\Rightarrow$ PC Binds

If $\phi_2 = \phi_H$, promised pensions must be raised further



Low Dependency Ratio $\Rightarrow$ PC Slack

If $\phi_3 = \phi_L$, promised pensions can be reset to p_j^0

